Bohdan Zelenskyi

ML Engineer | Backend Developer

AI Systems • Real-Time Processing • Production-Ready Services

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Profile

ML Engineer and Backend Developer with experience building real-time AI systems processing thousands of images daily and integrating ML into production environments.

Experience

Backend Developer | Sayorama (Startup)
2025–Present

- Built backend services for a real-time messaging platform across iOS, Android, and Web, inspired by Telegram.
- Designed and implemented low-latency communication with WebSockets, enabling scalable real-time communication.
- Developed async and event-driven backend flows to improve performance under concurrent load.
- Implemented caching and messaging mechanisms with Redis and queues, improving responsiveness and scalability.
- Built authentication, secrets management, and token rotation systems for secure session handling.
- Containerized services with Docker and Docker Compose and designed APIs for messaging, users, and synchronization.

Machine Learning / Computer Vision Engineer | KP MKPV KhMR (Production Systems)
2022–2025

- Built and deployed ML and computer vision pipelines processing thousands of images daily, reducing manual work by 60–80%.
- Developed and productionized OCR, object detection, face recognition, and license plate recognition models.
- Designed backend APIs with FastAPI and Flask, integrating ML services into business workflows.
- Optimized inference pipelines to achieve near real-time performance in production.
- Deployed services on Linux with Docker and Docker Compose, ensuring reproducibility and scalability.

Fullstack / Backend Developer | Wifi&Bed Co (Contract)
2023–2024

- Built CRM and analytics backend systems using Python, Flask, and MySQL.
- Designed data aggregation and analytics pipelines to support data-driven operations.
- Optimized SQL queries and backend performance, reducing latency and improving scalability.
- Designed system architecture prepared for future ML integration.

Selected Project

VisionInsight — AI Video Analytics & Multimodal Intelligence System
https://github.com/DrBarga/visioninsight

- Developed a real-time AI system for video and audio analysis capable of detecting people, objects, emotions, clothing attributes, and behavior.
- Built a multimodal pipeline combining computer vision with speech-to-text to transform raw data into structured insights.
- Implemented real-time video stream processing for live monitoring and analysis.
- Generated analytics outputs including charts, summaries, and insights for decision support.
- Designed a modular architecture deployed with Docker Compose for scalable orchestration.

Tech Stack

- **Machine Learning & AI:** Python, PyTorch, TensorFlow, Scikit-learn, speech-to-text, multimodal systems.
- **Computer Vision:** Object detection, OCR, face recognition, emotion recognition.
- **Backend & Systems:** FastAPI, Flask, REST APIs, WebSockets, async processing, HTTP / HTTPS.
- **Infrastructure & DevOps:** Docker, Docker Compose, Linux.
- **Data & Databases:** MySQL, SQL, Redis (caching, queues), Pandas, NumPy.
- **Testing & Tools:** Pytest, Git.

Education

M.Sc. in Computer Science (in progress)
National Technical University "KhPI"
Focus: Machine Learning, AI, Data Science

B.Sc. in Computer Science
National Technical University "KhPI"
Graduated

Languages

- English — B2
- German — B1 (intermediate, improving)